Grouped Bidirectional LSTM Network and Multistage Fusion Convolutional Transformer for Hyperspectral Image Classification

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***Abstract*— The efficiently and effectively discriminative spectral–spatial feature representation is essential for hyperspectral image (HSI) classification. However, most of the existing methods rely on patch-based convolutional neural networks (CNNs) whose ability to extract the global spatial information is very limited. To address this issue, in this article, we propose a two-branch network consisting of a grouped bidirectional long short-term memory (GBiLSTM) network and multistage fusion convolutional transformer (MFCT) for HSI classification. In the proposed GBiLSTM-MFCT, to extract the spectral features of HSI efficiently, a GBiLSTM network is designed by dividing the sequence features and hidden units of the BiLSTM network into several separate groups. To simultaneously extract the global and local spatial features of HSI, an MFCT is proposed by fusing the features of different levels obtained from the multiple phases of the convolutional vision transformer (CVT). Moreover, in the multiheaded attention module of each stage, the blueprint separable convolution-based self-attention (BSCA) module is designed, which is able to model the global and local spatial information effectively. The outputs of the GBiLSTM network and MFCT are fused to generate discriminative and robust spectral–spatial features for HSI classification. Experiments on three benchmark datasets of Indian Pine (IN), University of Pavia (UP), and Kennedy Space Center (KSC) demonstrate that the proposed GBiLSTM-MFCT exhibits higher classification performance with very limited labeled samples than eight state-of-the-art methods.**

***Index Terms*— Bidirectional long short-term memory (LSTM), convolutional vision transformer (CVT), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL image (HSI) consists of an abundant spectrum, which is collected from the electromagnetic radiation emitted, or reflected, by the observed target and spatial contextual information [1]. This enables the HSIs to offer a precise classification of land cover materials. HSIs' classification is to assign a specific category to each pixel using the spectral and spatial information, which has a variety of applications, such as environment management [2], anomaly detection [3], urban development [4], and national defense security [5]. However, owing to the high complexity of data and the very small number of labeled samples, the precise classification of HSI is still challenging.

Over the past decade, deep-learning (DL)-based HSI classification methods, such as autoencoders (AEs) [6], deep belief networks (DBNs) [7], recurrent neural networks (RNNs) [8], convolutional neural networks (CNNs) [9], [10], [11], [12], [13], generative adversarial networks (GANs) [14], capsule networks (CapsNets) [15], and graph convolutional networks (GCNs) [16], have been revealed significant superiorities over the traditional machine learning (ML) methods that rely on the handcrafted features and prior expert knowledge. Among the DL methods, CNNs that can extract the spectral features with spatial–contextual information by a series of hierarchical filters are popular tools for HSI classification. According to the input information of the network, CNN models for HSI classification include spectral CNNs, spatial CNNs, and spectral–spatial CNNs. For example, Wei *et al.* [17] proposed to process the spectral pixel by stacking multiple 1-D CNN layers, which only utilize the spectral information of a single pixel of HSI data for classification. Zhao and Du [18] and Liang and Li [19] proposed 2-D CNN-based models to extract the spatial features contained in the principal components of HSI data by principal component analysis (PCA). To better extract the spectral and spatial features simultaneously, Chen *et al.* [20] proposed a deep network with stacked 3-D CNN layers to effectively extract integrated spectral–spatial features for HSI classification. He *et al.* [21] explored the fusion of multiscale spectral–spatial features extracted by 3-D CNN layers with different convolution kernels for HSI classification.

With the CNN-based deep networks going deeper, the degradation problem exposes that, in this regard, the residual connectivity has been proposed to mitigate the data degradation for HSI classification by feeding shallow information to the deeper units. For example, Zhong *et al.* [22] proposed a 3-D spectral–spatial residual network (SSRN) that uses two connected spectral and spatial residual blocks to learn the spectral and spatial features of HSI. Paoletti *et al.* [23] proposed a pyramidal ResNet model that gradually increases the feature mapping dimension in all residual cells to obtain the

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robust spectral–spatial features of HSIs. Based on SSRN [22], Zhu *et al.* [24] developed spectral attention and spatial attention blocks to focus on useful information and suppress useless information for HSI classification. Roy *et al.* [25] proposed an improved ResNet to extract the joint spectral–spatial features for HSI classification where an attention-based adaptive spectral–spatial kernel module is developed to learn selective 3-D convolutional kernels.

To overcome the challenge of the limited training samples in HSI data versus the sufficient labeled samples required by the DL methods, many researchers have studied the HSI classification methods with very limited samples [26], [27], [28], [29], [30], [31], [32]. For example, Liu *et al.* [26] proposed a deep few-shot learning (DFSL) method for HSI classification, which learns a metric space by training a deep residual network based on 3-D CNN to make similar samples be close to each other. Li *et al.* [27] proposed a new deep cross-domain few-shot learning (DCFSL) method to realize HSI cross-domain few-shot classification, which is not easily to be affected by domain shift. The DCFSL method achieves better performance than DFSL with a few labeled samples. In addition, to the same purpose, a number of HSI classification methods based on transfer learning (TL) and active learning (AL) were proposed [28], [29], [30], [31], [32]. Xie *et al.* [29] proposed a superpixel pooling CNN with TL for HSI classification, which uses the TL strategy to improve the training efficiency. To address the poor performance produced by the training and testing images from different domains, Qu *et al.* [30] proposed a new TL method for HSI classification, which projects the HSI data from the source and target domains into a shared space based on their characteristics. In [31], a unified framework composed of AL and CNN was proposed for HSI classification where the CNN was trained with a limited number of labeled pixels, and the most informative pixels for labeling were actively selected. In [32], a multiview spatial–spectral AL (MVSS-AL) model was proposed for HSI classification, which designs a multiview leave-one-class-out (LOCO) strategy to explore more useful information in the limited training data.

However, in fact, simply stacking the standard 3-D CNN for simultaneously extracting the spectral and spatial features of HSI data results in high computational complexity and heavy redundancy in parameters, which is a disadvantage for HSI classification. To alleviate the problems, Roy *et al.* [13] proposed a hybird spectral–spatial network model composed of 3-D CNN and 2-D CNN layers for HSI classification, in which the 3-D CNN learns the spectral–spatial features and 2-D CNN learns the spatial features of HSI data. Xu *et al.* [33] proposed to employ the octave 3-D CNN for factorizing the feature maps by their frequency, instead of using the standard 3-D CNN, to reduce the spatial redundancy and enlarge the receptive field (RF) for HSI classification. Cui *et al.* [34] proposed a new 3-D deep network architecture for HSI classification, which decomposes the standard convolution into depthwise convolution and pointwise convolution to achieve high classification performance with very few parameters and low computational cost. Moreover, considering the high complementarity of the rich spectral and spatial information, another feasible solution is attempting to extract spectral and spatial features separately via a two-branch network, which can fully combine spectral and spatial information [35], [36], [37], [38], [39]. For example, Yang *et al.* [35] and Xu *et al.* [36] both proposed a two-branch architecture that uses a 1-D CNN subnetwork for extracting spectral features and a 2-D CNN subnetwork for extracting spatial features, respectively. Hao *et al.* [37] proposed a dual-stream architecture employing a stacked denoising self-encoder and a 2-D CNN-based network to extract the spectral and spatial features, respectively. These methods all use a few principal components as the inputs of the 2-D CNN subnetworks for spatial feature extraction, which can significantly reduce the computational consumption compared to the normal 3-D CNN-based methods. However, this type of 2-D CNN-based subnetwork only simply stacks the 2-D CNN layers, which easily causes the loss of spatial information. To improve the characterization capability of CNNs and enable the network to focus on more useful spectral bands and spatial locations, Hang *et al.* [38] proposed a two-branch dual-attention mechanism network based on 2-D CNN. It applies the channel attention and spatial attention blocks on the two branches of extracting spectral and spatial features, respectively, to further capture the refined features. However, the CNNs are not suitable for handling the spectral continuity of HSI data between adjacent bands. To this end, Xu *et al.* [39] proposed an end-to-end network for HSI classification, in which the spectral features are extracted by a band grouping-based long short-term memory (LSTM) network, and the spatial features are extracted through multiscale CNN subnetwork simultaneously. In this article, we also leverage a two-branch structure for extracting the spectral–spatial information of HSI for HSI classification.

Recently, the transformer models [40], which can handle sequential data and capture the long-term dependence of the data better than CNNs, have been introduced to HSI classification task [41], [42]. He *et al.* [41] first extracted the spatial information of each spectral band using a CNN and then proposed a modified transformer to encode the spectral dependencies between spectral bands. Hong *et al.* [42] proposed a new backbone network named SpectralFormer for HSI classification, which learns local spectral information in the neighboring spectral bands and reduces the loss of possible information by cross-layer skip connection. In spite of this, it should be noted that the computational complexity of each layer in the standard transformer is mainly controlled by the multiheaded self-attention layer and the feed-forward layer, requiring an overall $\mathcal{O}(n^2d + nd^2)$, whereas the bidirectional LSTM (BiLSTM) yields a complexity of $\mathcal{O}(nd^2)$, where n is the sequence length and d is the hidden feature dimension. It is obvious that the training cost of the standard transformer is higher than that of BiLSTM. Moreover, the BiLSTM still has a good generalization ability for learning correlation between sequence data [43]. Although the transformer can capture long-range (global) sequence information, it loses the ability to characterize the local contextual information. The recently proposed convolutional vision transformer (CVT) [44], which has a multilevel hierarchy similar to CNN, integrates the characteristics of CNN into the vision transformer (ViT) [45]

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architecture. The CVT has the ability to learn global and local spatial information. However, the employed convolutional projection mapping operation easily leads to information loss when extracting the local spatial information. In addition, the shallow information features are often neglected in the multistage structure of CVT.

To solve the aforementioned issues, in this article, we propose a two-branch network, the grouped BiLSTM network and multistage fusion convolutional transformer (GBiLSTM-MFCT), for HSI classification. The GBiLSTM-MFCT exploits the BiLSTM and CVT as the baselines to efficiently and effectively learn the spectral feature of a single pixel, and the local and global spatial correlations of HSI. To characterize the strong contiguity among the spectral bands of HSI efficiently, we propose the GBiLSTM network that groups the sequence features and hidden units and, consequently, further reduces the training parameters of the BiLSTM network. For the spatial feature extraction, we propose the MFCT that attempts to fuse the features obtained by the multiple stages. In each stage, to enlarge the perceptual field, a double convolutional embedding (DCE) is designed, which forms a convolutional projection embedding layer. Meanwhile, to reduce the high redundancy of the CNN kernel along the depth direction and allow the self-attention module to further model local spatial contextual information, the blueprint separable convolution (BSConv) is adopted to implement the convolutional projection in the self-attention module and form a BSConv-based self-attention (BSCA) module. Considering the correlation and complementarity of features obtained by the different stages in MFCT, the features of different levels are fused to enhance the feature representation. Finally, to make full use of the complementarity between spectral and spatial features, the outputs of the GBiLSTM network and MFCT are fused. The proposed GBiLSTM-MFCT can effectively balance the tradeoff between the model performance and complexity. The main contributions of this article are summarized as follows.

1) We propose a novel end-to-end two-branch network for HSI classification that consists of the GBiLSTM network for spectral feature extraction and the MFCT for spatial feature extraction. The proposed GBiLSTM-MFCT can simultaneously capture the interspectral interactions and spatial sequences of long-range interactions.
2) We propose the GBiLSTM network that divides the feature vectors of the input sequences and BiLSTM hidden units into several separate groups and then fuses the features extracted from different groups. The proposed GBiLSTM network can highly reduce the number of parameters required and extract the spectral feature of HSI efficiently.
3) To effectively extract the global and local spatial features of HSI, the MFCT is proposed where the features of different levels obtained from the multiple phases of the CVT are fused. In each stage of MFCT, the DCE and BSCA modules are designed which can improve the classification performance significantly.

The remainder of this article is organized as follows. In Section II, we briefly introduce bidirectional long-short term memory networks and CVTs. In Section III, we give the detailed design of the proposed GBiLSTM-MFCT for HSI classification. The experimental results and discussions are presented in Section IV. Finally, Section V concludes this article.

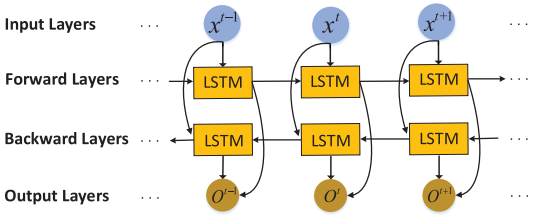


Fig. 1. Architecture of a BiLSTM network.

II. Related Work

A. Bidirectional Long-Short Term Memory Network

RNN [46] is a neural network with short-term memory function. The output of its hidden unit can obtain the information passed by other neurons and the information passed by itself. RNNs have shown good performance in many sequence data applications [47], [48]. However, in dealing with long sequences, RNNs have gradient explosion and vanishing problems. The LSTM network [49] uses cell states and three kinds of gates, the input gate, the forget gate, and the output gate, to keep and control information, which can effectively solve the gradient explosion and vanishing problems. As the LSTM network accesses information in one direction (via forward states), therefore, the BiLSTM network was proposed to encode information both via forward states and backward states, which can capture the deep relationship between sequences. The architecture of the BiLSTM network is shown in Fig. 1. The final output of the BiLSTM network consists of the outputs of the forward and backward layers at the same time step. To fully explore the strong contiguity among the spectral bands of HSI, we propose an efficient GBiLSTM network for spectral feature extraction.

B. Convolutional Vision Transformers

Transformers [40], which completely rely on a self-attention mechanism to capture the global dependencies, were proposed for natural language processing [50]. Recently, transformer-based architectures have been proposed to replace CNNs for image classification [45], target detection [51], instance segmentation [52], and so on. The success of transformers refers to a large extent from the multiheaded self-attention mechanism. Compared with CNNs, which can only capture the local information by nonlinear filters, transformers can perform global information extraction on the input image through the self-attention mechanism.

The ViT [45] was introduced to extend the standard transformer to image classification and almost utilizes the encoder module of the transformer for feature extraction. Using ViT for image classification, the image is first segmented into multiple image patches, and a linear embedding sequence of these image patches is used as the input of the transformer encoder. Recently, several transformer-based architectures have been proposed to address the image understanding

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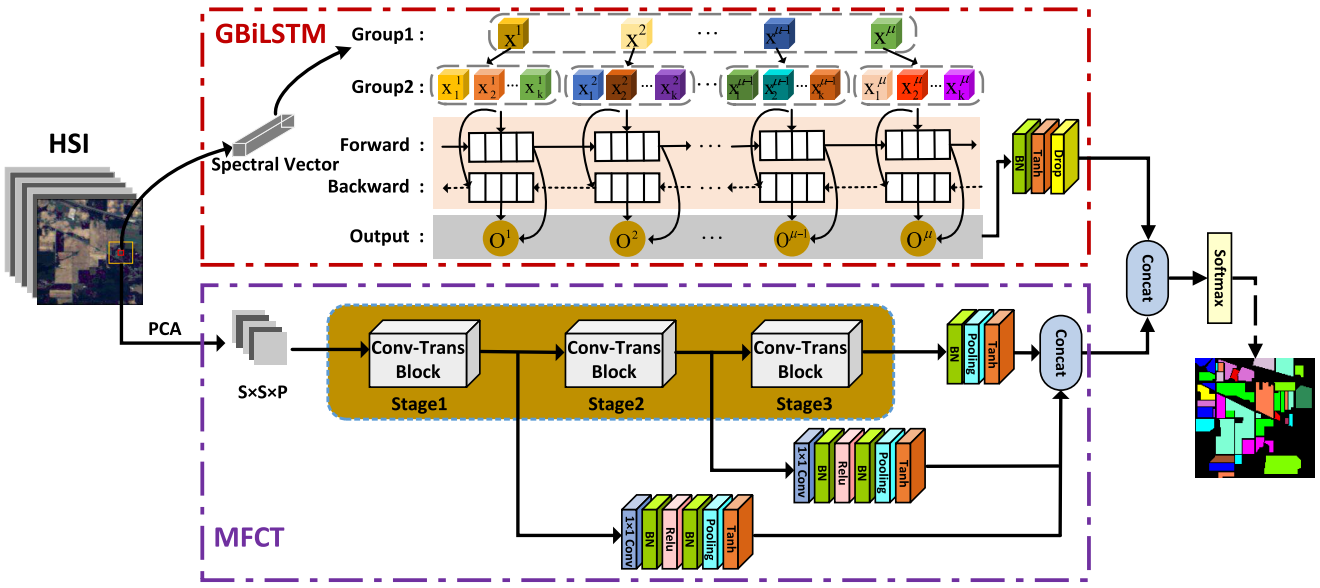


Fig. 2. Architecture of the proposed GBiLSTM-MFCT that consists of a GBiLSTM branch and an MFCT branch. The final extracted spectral features and spatial features are fused and fed to the softmax layer for classification. The Conv-Trans Block in the MFCT represents a Conv-Trans Block.

tasks and shown competitive performance [44], [53], [54], [55], [56]. For example, an image transformer model with a distillation strategy based on a distillation token was developed and trained efficiently for image classification [53]. In [55], a pyramidal vision transformer (PVT), which has a progressive shrinking pyramid and spatial-reduction attention for reducing the computation consumption of the transformer, was proposed and verified as more suited for image segmentation and object detection. However, the deficiency of the visual transformers is that they cannot capture the local spatial structure of images, whereas the CNNs can compensate. In consideration of this, to capture both the local and global spatial information, Wu *et al.* [44] proposed the CVT by introducing CNN into the transformer, which uses convolutional projection operation instead of linear projection in the embedding layer and the transformer encoder. Therefore, we design an MFCT for spatial feature extraction of HSI data.

III. Methodology

The proposed GBiLSTM-MFCT model for HSI classification consists of two branches: the GBiLSTM network for spectral feature extraction and MFCT for spatial feature extraction. The architecture is shown in Fig. 2. Let $\boldsymbol{\chi} \in \mathbb{R}^{H\times W\times B}$ denote the original hyperspectral data, where H and W are the height and width of the spatial dimensions, and B is the number of spectral bands. The input of the GBiLSTM network for spectral feature extraction is a single pixel vector $\boldsymbol{x} \in \mathbb{R}^{1\times B}$. Due to the high intraclass variability and interclass similarity of HSI data, for the spatial feature extraction by MFCT, we reduce the redundancy of the spectral and retain sufficient spatial information. The raw HSI data are preprocessed by the PCA. The number of spectral bands is reduced from B to P, and the spatial dimension retains unchanged. The 3-D patch cube $\boldsymbol{z} \in \mathbb{R}^{S\times S\times P}$ that is centered at each labeled pixel with the neighboring regions is the input of the proposed MFCT.

A. Grouped Bidirectional LSTM Network

The LSTM network has been used for HSI classification as it can utilize the contextual information of neighboring data to extract robust and invariant spectral features of HSIs. Compared with the LSTM network, the BiLSTM network can make full use of the information of sequences via forward and backward states to train the parameters more effectively. Inspired by this, to effectively reduce the number of parameters and accelerate the training for the spectral feature extraction, we construct a new network architecture (i.e., GBiLSTM), which has two grouping phases, as shown in Fig. 2.

In the conventional LSTM-based spectral extraction methods, the spectral sequence is divided into individual bands and fed to the network one by one, that is, input one band at a time step. This brings about too few input features of each sequence to train and leads to a too deep network as there are usually hundreds of spectral bands in HSI. Therefore, we implement an initial grouping strategy for each spectral pixel vector [39]. Given a spectral vector of $\boldsymbol{x} = [\boldsymbol{s}_1, \ldots, \boldsymbol{s}_t, \ldots, \boldsymbol{s}_B]$, where $\boldsymbol{s}_t$ is the tth band, we divide $\boldsymbol{x}$ into μ sequences, denoted by $\boldsymbol{x}_g = \{\boldsymbol{x}^1, \ldots, \boldsymbol{x}^t, \ldots, \boldsymbol{x}^\mu\}$, where $\boldsymbol{x}^t = [\boldsymbol{s}_t, \boldsymbol{s}_{t+\mu}, \ldots, \boldsymbol{s}_{t+\mu(m-1)}]$ is the input sequence at the tth time step, and the feature dimension of each sequence $\boldsymbol{x}^t$ is defined as $m = \lfloor B/\mu \rfloor$.

To reduce the number of parameters and accelerate the training, inspired by the LSTM factorization technique [57], we propose to divide the sequence $\boldsymbol{x}^t, t = 1, 2, \ldots, \mu$ into independent groups. For convenience, $\boldsymbol{x}^t$ is rewritten as $\boldsymbol{x}^t = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_i, \ldots, \boldsymbol{v}_m]$, where $\boldsymbol{v}_i = \boldsymbol{s}_{t+\mu(i-1)}$, $\boldsymbol{x}^t$ is divided into k independent groups, i.e., $\boldsymbol{x}^t = \{\boldsymbol{x}_1^t, \ldots, \boldsymbol{x}_i^t, \ldots, \boldsymbol{x}_k^t\}$, where $\boldsymbol{x}_i^t = [\boldsymbol{v}_i, \boldsymbol{v}_{i+1}, \ldots, \boldsymbol{v}_{i+l-1}]$, and the feature dimension of each independent sequence group is $l = \lfloor \mu/k \rfloor$. Then, the forward

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hidden layer unit $\boldsymbol{h}^t$ and the backward hidden layer unit $\tilde{\boldsymbol{h}}^t$ $(t = 1, \ldots, \mu)$ are accordingly divided into k independent groups, respectively, i.e., $\boldsymbol{h}^t = \{\boldsymbol{h}_1^t, \ldots, \boldsymbol{h}_i^t, \ldots, \boldsymbol{h}_k^t\}$, $\tilde{\boldsymbol{h}}^t = \{\tilde{\boldsymbol{h}}_1^t, \ldots, \tilde{\boldsymbol{h}}_i^t, \ldots, \tilde{\boldsymbol{h}}_k^t\}$. The output of each group at time step t is computed as follows:

$$\boldsymbol{h}_i^t = f\left(\overrightarrow{\boldsymbol{U}}\boldsymbol{x}_i^t + \overrightarrow{\boldsymbol{V}}\boldsymbol{h}_i^{t-1} + \overrightarrow{\boldsymbol{b}}\right) \tag{1}$$

$$\tilde{\boldsymbol{h}}_i^t = f\left(\overleftarrow{\boldsymbol{U}}\boldsymbol{x}_i^t + \overleftarrow{\boldsymbol{V}}\tilde{\boldsymbol{h}}_i^{t+1} + \overleftarrow{\boldsymbol{b}}\right) \tag{2}$$

$$\boldsymbol{O}_i^t = g\left(\boldsymbol{U}^o\boldsymbol{h}_i^t + \boldsymbol{V}^o\tilde{\boldsymbol{h}}_i^t + \boldsymbol{b}^o\right) \tag{3}$$

where $\overrightarrow{\boldsymbol{U}}$, $\overleftarrow{\boldsymbol{U}}$, $\boldsymbol{U}^o$, $\overrightarrow{\boldsymbol{V}}$, $\overleftarrow{\boldsymbol{V}}$, and $\boldsymbol{V}^o$ are the weight matrices; $\overrightarrow{\boldsymbol{b}}$, $\overleftarrow{\boldsymbol{b}}$, and $\boldsymbol{b}^o$ are the bias parameters; f is the nonlinear activation function of the hidden layer; and g is a concatenation function between the forward hidden state and backward hidden state. Finally, the output of each group at time step t is fused

$$\boldsymbol{O}^t = \left[\boldsymbol{O}_1^t, \boldsymbol{O}_2^t, \ldots, \boldsymbol{O}_k^t\right]. \tag{4}$$

The fused output is fed to a batch normalization layer and a Tanh activation layer. To prevent the network from overfitting, a dropout layer is added to obtain the spectral features.

It can be noted that each group in the second grouping phase of the GBiLSTM network is independent, and the architecture of the GBiLSTM network has high parallelism. Moreover, assuming that d_x and d_h are the dimensions of the input and hidden layers, the number of parameters of a layer in the standard BiLSTM is $8d_h(d_h + d_x + 1)$, while, in the proposed GBiLSTM network (with k groups in the second grouping phase), it can be reduced to $8(d_h/k)((d_h/k) + (d_x/k) + 1)$.

B. Multistage Fusion Convolutional Transformer

The CVTs destruct the internal structural information of the image in the process of patching the input image and the attention mechanism that focuses on long-range relationships and ignores the local spatial information of the image. Compared with the CVT, CNN can extract the local relationship of 2-D data. Therefore, in this section, to learn the deep discriminative local–global spatial feature of HSI, the robust MFCT is developed based on CVT.

In the MFCT, there are three stages, each of which is actually a convolutional transformer block (Conv-Trans Block). As shown in Fig. 3, a Conv-Trans Block consists of two parts: a convolutional projection embedding layer and a convolutional transformer encoder. For a standard 2-D CNN, the ith output feature map of the $(l+1)$th convolutional layer with activation function is represented as

$$\boldsymbol{H}_i^{l+1} = \mathrm{ReLU}\left(\sum_j \left(\boldsymbol{H}_j^l * \boldsymbol{w}_i^{l+1}\right) + \boldsymbol{b}_i^{l+1}\right) \tag{5}$$

where $\boldsymbol{H}_j^l$ is the jth output feature map of the lth convolutional layer, $\boldsymbol{w}_i^{l+1}$ and $\boldsymbol{b}_i^{l+1}$ denote the ith convolution kernel and bias in the $(l+1)$th convolution layer, respectively, ReLU is the activation function ReLU, and $*$ represents the convolution operation.

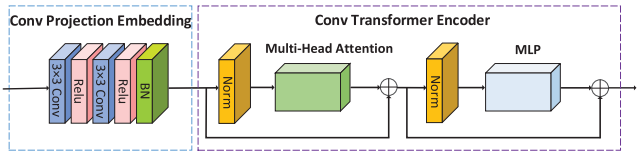


Fig. 3. Architecture of a Conv-Trans Block.

1) Convolutional Projection Embedding Layer: In the convolutional projection embedding layer, different from the original CVT model that adopts one 2-D CNN, we propose the DCE. Specifically, two consecutive convolution operations are performed in the embedding layer, where each convolution is a 2-D CNN with a convolution kernel of 3×3 and a padding of 0. After each convolution operation, the feature maps are activated by the ReLU activation function. Before feeding to the convolutional transformer encoder, batch normalization is used. The output of the convolutional projection embedding layer $\boldsymbol{H}_c$ is represented as

$$z^l = \mathrm{ReLU}(z^{l-1} * \boldsymbol{W}^l + \boldsymbol{b}^l) \tag{6}$$

$$\boldsymbol{H}_c = BN(z^2) \tag{7}$$

where $l = 1, 2$, z^0 is the input of the convolutional projection embedding layer, z^1 and z^2 are the output feature maps of the 1th and 2th convolutional layers, respectively, among which $z^2 \in \mathbb{R}^{D_i \times S_i \times S_i}$, D_i and $S_i \times S_i$ denote the number of channel and the spatial size of the output feature maps of the convolutional projection embedding layer in the ith stage of MFCT, BN is the batch standardization, and $\boldsymbol{W}^l$ and $\boldsymbol{b}^1$ are the convolution kernels and bias in the lth convolution layer. As the convolutional projection embedding layer can extract the local spatial information of HSI data, the designed DCE can obtain a larger perceptual field and a feature map with more global, semantic-level features.

2) Convolutional Transformer Encoder: The convolutional transformer encoder, which consists of a multiheaded self-attention layer and an MLP layer, is the key part of the proposed MFCT. In the multiheaded self-attention layer, there are multiple attention heads to focus on the key information of the feature maps from different aspects. In one attention head, by convolutionally projecting with BSConv of the output feature maps of the convolutional projection embedding layer and flattening into 1-D, the query tensor ($\boldsymbol{Q} \in \mathbb{R}^{D_m \times D_k}$), the key tensor ($\boldsymbol{K} \in \mathbb{R}^{D_n \times D_k}$), and the value tensor ($\boldsymbol{V} \in \mathbb{R}^{D_n \times D_k}$) are obtained. The output of one attention head $\boldsymbol{A} \in \mathbb{R}^{D_m \times D_k}$ is calculated as follows:

$$\boldsymbol{A} = \mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{D_k}}\right)\boldsymbol{V} \tag{8}$$

where $(1/\sqrt{D_k})$ is the normalization scaling factor and $softmax$ is the softmax function. Then, the outputs of multiple attention heads are concatenated and reshaped into the 2-D feature maps. Through one 1×1 2-D convolution, we can obtain the outputs whose dimension is equal to the dimension of the input of the multiheaded self-attention layer. The output feature maps of multihead attention $\hat{\boldsymbol{H}}_{\mathrm{msa}} \in \mathbb{R}^{D_i \times S_i \times S_i}$ are

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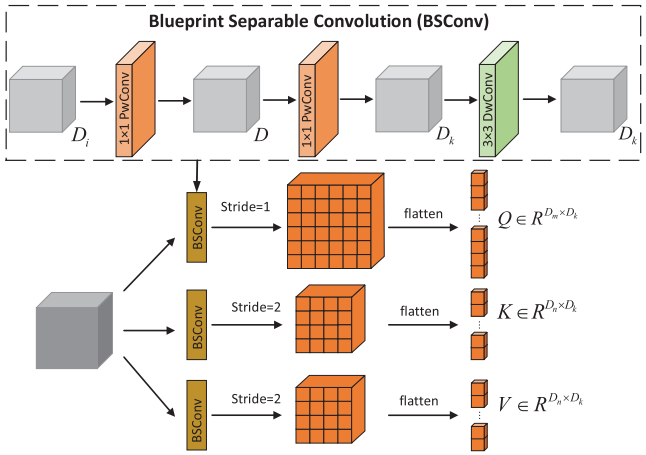


Fig. 4. Architecture of attention module convolutional projection with BSConv.

computed as follows:

$$\boldsymbol{H}_{\text{msa}} = [\boldsymbol{A}_1, \boldsymbol{A}_2, \ldots, \boldsymbol{A}_h] \tag{9}$$

$$\hat{\boldsymbol{H}}_{\text{msa}} = R(\boldsymbol{H}_{\text{msa}}) * \boldsymbol{W}_m + \boldsymbol{b}_m \tag{10}$$

where $\boldsymbol{A}_i$ is the output of the ith attentional head, $\boldsymbol{H}_{\text{msa}} \in \mathbb{R}^{D_m \times D_h}$, $(D_h = D_k \cdot h)$ is the combination of h attentional heads, R is the dimension reshaping operation, and $\boldsymbol{W}_m$ and $\boldsymbol{b}_m$ are convolution kernel and bias parameters, respectively.

The subsequent MLP layer consists of two 1×1 2-D convolutional layers and an ReLU activation layer. Both the attention and MLP layers in the transformer encoder use residual connectivity and layer normalization. Let $\boldsymbol{H}_{\text{mlp}} \in \mathbb{R}^{D_i \times S_i \times S_i}$ represent the input of the MLP layer; the output of the convolutional transformer encoder $\boldsymbol{H}_t \in \mathbb{R}^{D_i \times S_i \times S_i}$ is calculated by the following equations:

$$\boldsymbol{H}_{\text{mlp}} = \text{Norm}(\hat{\boldsymbol{H}}_{\text{msa}}) + \boldsymbol{H}_c \tag{11}$$

$$\hat{\boldsymbol{H}}_{\text{mlp}} = \text{ReLU}(\boldsymbol{H}_{\text{mlp}} * \boldsymbol{W}_1 + \boldsymbol{b}_1) * \boldsymbol{W}_2 + \boldsymbol{b}_2 \tag{12}$$

$$\boldsymbol{H}_t = \text{Norm}(\hat{\boldsymbol{H}}_{\text{mlp}}) + \boldsymbol{H}_{\text{mlp}} \tag{13}$$

where Norm represents the layer normalization operation, $\boldsymbol{W}_1$, $\boldsymbol{W}_2$ and $\boldsymbol{b}_1$, $\boldsymbol{b}_2$ are the 1×1 convolution kernels and bias parameters in the convolution layers, respectively.

3) Convolutional Projection With BSConv of Attention Module: In the multiheaded attention layer of the transformer encoder, we propose to employ the BSConv [58] to achieve the convolutional projection and construct the BSCA module for modeling the global and local spatial information effectively because the BSConv reduces the intrakernel redundancy by representing each filter kernel with a 2-D blueprint and assigning weights in the depth direction. It can separate the standard convolution efficiently. The structure of BSConv and the convolutional projection is illustrated in Fig. 4.

A BSConv block contains D_k filter kernels $\boldsymbol{F}_m^{(n)} \in \mathbb{R}^{D_i \times 3 \times 3}$, where each filter kernel is defined as a blueprint $\boldsymbol{B}^{(n)} \in \mathbb{R}^{1 \times 3 \times 3}$ multiplied with the corresponding weights $\boldsymbol{w}_{n,m}$. $\boldsymbol{F}_m^{(n)}$ is expressed as

$$\boldsymbol{F}_m^{(n)} = \boldsymbol{w}_{n,m} \cdot \boldsymbol{B}^{(n)} \tag{14}$$

where $m \in \{1, 2, \ldots, D_i\}$ and $n \in \{1, 2, \ldots, D_k\}$. By decomposing matrix $\boldsymbol{W} = (\boldsymbol{w}_{n,m})$, we get $\boldsymbol{W}' = (\boldsymbol{w}'_{n,d})$ and $\boldsymbol{W}'' = (\boldsymbol{w}''_{d,m})$, i.e., $\boldsymbol{W} = \boldsymbol{W}'.\boldsymbol{W}''$, $\boldsymbol{W} \in \mathbb{R}^{D_k \times D_i}$, $\boldsymbol{W}' \in \mathbb{R}^{D_k \times D}$, $\boldsymbol{W}'' \in \mathbb{R}^{D \times D_i}$, and $D = 2D_i$. The matrix $\boldsymbol{W}''$ can be viewed as the set of D basis vectors, while $\boldsymbol{W}'$ is a subspace version of $\boldsymbol{W}$. The output $\boldsymbol{H}_{q/k/v}$ of the projection layer can be deduced from the following equation:

$$\boldsymbol{H}_{q/k/v} = \left(\sum_{m=1}^{D_i} \boldsymbol{H}_c * \left(\sum_{d=1}^{D} \boldsymbol{w}'_{n,d} \cdot \boldsymbol{w}''_{d,m} \right) \cdot \boldsymbol{B}^{(n)} \right) \tag{15}$$

where $\boldsymbol{w}_{n,m} = \sum_{d=1}^{D} \boldsymbol{w}'_{n,d} \cdot \boldsymbol{w}''_{d,m}$. Due to each filter blueprint being independent of the input channel, the above equation can be rewritten as

$$\boldsymbol{H}_{q/k/v} = \left(\sum_{d=1}^{D} \left(\sum_{m=1}^{D_i} \boldsymbol{H}_c \cdot \boldsymbol{w}''_{d,m} \right) \cdot \boldsymbol{w}'_{n,d} * \boldsymbol{B}^{(n)} \right). \tag{16}$$

For implementation, the BSConv process is to perform two 1×1 pointwise convolutions of the input data $\boldsymbol{H}_c$, whose kernel numbers are D and D_k, and one 3×3 depthwise convolution, whose kernel number is D_k. The two pointwise convolutions can also be viewed as one 1×1 pointwise convolution decomposition. Thus, the matrices $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ used to obtain the attention scores can be represented as

$$\boldsymbol{Q}/\boldsymbol{K}/\boldsymbol{V} = F_{\text{fla}}(\text{BSConv}(\boldsymbol{H}_c)) \tag{17}$$

where BSConv consists of two pointwise convolutions and one depthwise convolution, and F_{fla} represents reshaping into 1-D.

To sparse the weight matrices, we constrain the basis $\boldsymbol{W}''$ being orthogonal. The regularization loss is defined as

$$L_0 = \left\| \boldsymbol{W}'' \boldsymbol{W}''^T - \boldsymbol{I} \right\|_F \tag{18}$$

where $\boldsymbol{I}$ is an identity matrix and $\| \cdot \|_F$ is the Frobenius norm of a matrix. This loss can be added to the subsequent classification loss to train the whole model.

4) Multistage Fusion: Through the multistage CVT, whose structure is similar to the hierarchical architecture of CNNs, multiple-level feature maps with different resolutions are obtained. Considering that the features obtained in different stages are complementary and strongly correlated, we propose to obtain the spatial features by adaptive averaging pooling and fusion of the feature maps obtained from the three stages. The batch normalization and Tanh activation function processing are used before and after the pooling operation. To increase the richness of the features, before feature fusion, by performing 1×1 2-D convolution operations on the first two stages, the number of feature maps of the first two stages becomes the same as that of the third stage. A batch normalization and ReLU activation are performed before and after the convolution operation. The fused spatial features can be calculated as follows:

$$\hat{\boldsymbol{H}}_t^i = \text{ReLU}\left(BN\left(\boldsymbol{H}_t^i * \boldsymbol{W}_t^i + \boldsymbol{b}_t^i\right)\right) \tag{19}$$

$$\boldsymbol{O}_m = \left[F_{\text{bat}}\left(\hat{\boldsymbol{H}}_t^1\right), F_{\text{bat}}\left(\hat{\boldsymbol{H}}_t^2\right), F_{\text{bat}}\left(\boldsymbol{H}_t^3\right) \right] \tag{20}$$

where $i = \{1, 2\}$, $\boldsymbol{H}_t^1$, $\boldsymbol{H}_t^2$, and $\boldsymbol{H}_t^3$ are the output feature maps of the three-stage transformer encoder of MFCT, respectively, $\boldsymbol{W}_t^i$ and $\boldsymbol{b}_t^i$ are the convolution kernels and bias in the 1×1 2-D convolution layer, and F_{bat} denotes a series of operations, including batch normalization, adaptive average pooling, Tanh activation, and reshaping to 1-D.

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C. Feature Fusion and Classification

According to the existing HSI classification methods, we can find that the spatial features of HSIs are more effective for pixel classification than the spectral features, but the classification using spatial features alone is prone to misclassify some pixels with large intraclass differences. The fusion of complementary spectral and spatial features for classification can effectively avoid such a situation. As the proposed two networks, respectively, focus on the extraction of spectral and spatial information of HSI, therefore, we fuse them by

$$\boldsymbol{O} = [\boldsymbol{O}_l, \boldsymbol{O}_m] \tag{21}$$

where $\boldsymbol{O}$ denotes the final fusion result, $\boldsymbol{O}_l$ denotes the spectral features extracted by the GBiLSTM network, and $\boldsymbol{O}_m$ denotes the spatial features extracted by MFCT. Finally, a fully connected layer with a softmax function is used for classification.

To train the proposed model, we choose the cross-entropy function L_c as the class loss function, and the regularized loss L_o obtained in the convolutional projection is summed up with the class loss function by a weighting factor α. The complete loss function of the GBiLSTM-MFCT is defined as

$$\begin{aligned} L &= L_c + \alpha L_o \\ &= -\frac{1}{M}\sum_{i=1}^{M}\sum_{j=1}^{C}\mathbf{y}_i^j \log \hat{\mathbf{y}}_i^j + \alpha\left\|\mathbf{W}''\mathbf{W}''^T - \mathbf{I}\right\|_F \end{aligned} \tag{22}$$

where C is the number of categories, M is the size of the training batch, α is the weighting factor, and $\mathbf{y}$ and $\hat{\mathbf{y}}$ are the true value and predicted labels, respectively.

IV. Experiments and Analysis

In this section, a series of intensive experiments are conducted on three HSI datasets to evaluate the performance of the proposed method.

A. Experimental Datasets

1) Indian Pine (IN) Dataset: This dataset was the first test data used for HSI classification and was imaged by the Airborne Visual Infrared Imaging Spectrometer (AVIRIS) in 1992 on a piece of IN tree in Indiana, USA. We generally use the remaining 200 spectral bands excluding the 20 bands that cannot be reflected by water, which have a wavelength range of 0.4–2.5μm, a spatial resolution of about 20 m, and a spatial size of 145 $\times$ 145. The ground truth contains 10 249 available pixels and 16 plant classes.

2) University of Pavia (UP) Dataset: This dataset is a part of the hyperspectral data of the images made by the German airborne reflectance optical spectral imager of the city of Pavia, Italy, in 2003. Twelve of the bands were removed due to noise effects, and we used the remaining 103 spectral bands. It has a wavelength range of 0.43–0.86 μm, a spatial resolution of 1.3 m, and a spatial size of 610 $\times$ 340, and contains 42 776 available pixels. Moreover, the image contains nine land cover classes.

3) Kennedy Space Center (KSC) Dataset: This dataset was taken by the AVIRIS sensor at KSC in Florida on March 23, 1996, and it contains 224 bands. It has 176 bands after removing the water vapor noise bands, a spatial resolution of 18 m, and a spatial size of 512 $\times$ 614. The dataset contains 5211 available pixels and 13 upland and wetland classes.

B. Experimental Setup

To verify the effectiveness of the proposed method, we compare the proposed GBiLSTM-MFCT with eight state-of-the-art deep methods. The compared methods include 3DCNN [12], M3DCNN [21], SSUN [39], SSRN [22], HybridSN [13], MSOctCaps [15], A^2S^2K-ResNet [25], and SpectralFormer [42]. Their brief descriptions and experimental configurations are given as follows.

1) *3DCNN:* For the 3DCNN, the network architecture consists of four 3-D convolutional blocks and fully connected layers; the first two convolutional blocks include a 3-D convolutional layer with a convolutional kernel of 3 $\times$ 3 $\times$ 3, an ReLU activation layer, and a 3-D convolutional layer with a convolutional kernel of 3 $\times$ 1 $\times$ 1 (as a pooling layer). The latter two convolutional blocks include a 3-D convolutional layer with a convolutional kernel of 3 $\times$ 1 $\times$ 1 and an ReLU activation layer. The implementation process is referred to as the 3-DCNN in [12].
2) *M3DCNN:* For M3DCNN, we use the same network architecture as in [21], and we fix the learning rate to $5e-4$ and epochs to 200.
3) *SSUN:* For the DSCNN branch in SSUN, we set the input size to 25 $\times$ 25 and keep the first five principal components. The configuration and implementation of the model are referred to [39].
4) *SSRN:* It is implemented exactly the same as the original paper [22].
5) *HybridSN:* It is a network architecture with a mixture of 2-D and 3-D convolutional layers. We implemented HybridSN using the same hyperparameters as in [13].
6) *MSOctCaps:* The experimental configurations are the same as the original paper [15].
7) *A^2S^2K-ResNet:* The details of the implementation can be referred to [25].
8) *SpectralFormer:* The experimental configurations are the same as the original paper on the IN and UP datasets [42]. On the KSC dataset, the training epochs for the pixelwise and patchwise inputs are set to 800 and 480, respectively; the other experimental configurations are set as the UP dataset.

The learning rate of our proposed model is set to $3e-4$. In the GBiLSTM network, μ is set to 3, k is set to 2, the dropout rate is fixed to 0.5, and the number of hidden layer units is set to 256. In the MFCT network, the input size is set to 25 $\times$ 25 $\times$ 5. The network architecture configuration of the proposed MFCT is described in detail in Table I. Here, Head_i, Depth_i, and R_i denote the number of attentional heads, depth, and feature size expansion ratio of MLP layers in phase i of MFCT, respectively. Our proposed models are all trained with a batch size of 128 for 200 epochs. The weighting factor

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TABLE I
PROPOSED MFCT NETWORK ARCHITECTURE CONFIGURATION

	Layname	Configuration	Outsize
Stage1	Conv Embed	Conv1 32×3×3 stride=1 Conv2 32×3×3 stride=1	32×21×21
	Conv Proj	Q Conv 64×3×3 stride=1 K,V Conv 64×3×3 stride=2	
	MHSA	$Head_1$=2 $Depth_1$=2	
	MLP	R_1=2	
Stage2	Conv Embed	Conv1 64×3×3 stride=1 Conv2 64×3×3 stride=1	64×17×17
	Conv Proj	Q Conv 64×3×3 stride=1 K,V Conv 64×3×3 stride=2	
	MHSA	$Head_2$=3 $Depth_2$=2	
	MLP	R_2=2	
Stage3	Conv Embed	Conv1 128×3×3 stride=1 Conv2 128×3×3 stride=2	128×7×7
	Conv Proj	Q Conv 64×3×3 stride=1 K,V Conv 64×3×3 stride=2	
	MHSA	$Head_3$=4 $Depth_3$=2	
	MLP	R_3=2	

TABLE II
NUMBERS OF TRAINING AND TEST PIXELS FOR THE IN DATASET

No.	Class Name	Train	Val	Test
1	Alfalfa	3	3	40
2	Corn-notill	43	43	1342
3	Corn-mintill	25	25	780
4	Corn	8	8	221
5	Grass-pasture	15	15	453
6	Grass-trees	22	22	686
7	Grass-pasture	3	3	22
8	Hay-windrowed	15	15	448
9	Oats	3	3	14
10	Soybean-notill	30	30	912
11	Soybean-mintill	74	74	2307
12	Soybean-clean	18	18	557
13	Wheat	7	7	191
14	Woods	38	38	1189
15	Buildings-Grass	12	12	362
16	Stone-Steel	3	3	87
	Total	319	319	9611

TABLE III
NUMBERS OF TRAINING AND TEST PIXELS FOR THE UP DATASET

No.	Class Name	Train	Val	Test
1	Asphalt	67	67	6497
2	Meadows	187	187	18275
3	Gravel	21	21	2057
4	Trees	31	31	3002
5	Painted metal sheets	14	14	1317
6	Bare Soil	51	51	4927
7	Bitumen	14	14	1302
8	Self-Blocking Bricks	37	37	3608
9	Shadows	10	10	927
	Total	432	432	41912

α in the loss function is set to 0.1. Moreover, to evaluate the performance of the proposed models, we used the overall accuracy (OA), average accuracy (AA), accuracy per class, and kappa coefficient (κ) as the objective criteria. Tables II–IV present the sample sizes of the training, validation, and test sets of the three HSI datasets. For all datasets, we performed all experiments five times to calculate the corresponding AA and standard deviation. All experiments are implemented on a computer with an Intel Core i7-9700K CPU and a GeForce RTX 2080 graphics card.

TABLE IV
NUMBERS OF TRAINING AND TEST PIXELS FOR THE KSC DATASET

No.	Class Name	Train	Val	Test
1	Scrub	31	31	699
2	Willow swamp	10	10	223
3	Cabbage palm	11	11	234
4	Cabbage oak	11	11	230
5	Slash pine	7	7	147
6	Oak hammock	10	10	209
7	Hardwood swamp	5	5	95
8	Graminoid marsh	18	18	395
9	Spartina marsh	21	21	478
10	Cattail marsh	17	17	370
11	Salt marsh	17	17	385
12	Mud flats	21	21	461
13	Water	38	38	851
	Total	217	217	4777

C. Model Performance Comparison

1) Quantitative Comparisons: The first set of experiments is to provide the quantitative comparison of OA, AA, and κ classification scores of the current well-known methods on the three datasets, including the accuracy of each category for each dataset, as shown in Tables V–VII. From the tables, we can find that, for all the datasets, the SSUN, SSRN, A^2S^2K-ResNet, HybridSN, and MSOctCaps obtain higher classification accuracy than the 3DCNN and M3DCNN, which use only 3-D convolutional layers. This is because simply stacking 3-D convolutional layers generates a large amount of feature redundancy. In addition, A^2S^2K-ResNet has higher classification accuracy compared to SSRN on all datasets. This is mainly because A^2S^2K-ResNet can better encode spectral and spatial features by adjusting the convolutional RFs. For learning the spectral information of a single pixel of HSI, the proposed GBiLSTM provides high OAs than the SpectralFormer with pixelwise input and the spatial–spectral SpectralFormer with patchwise input on the three datasets. This may be due to that the proposed GBiLSTM can better cope with the problem of the high intraclass variation and interclass similarity of the spectrum. The proposed MFCT also has a powerful performance by focusing on the spatial alignment of HSI data to obtain a finer representation of spectral information. It can be observed from the tables that the proposed GBiLSTM-MFCT has higher OA, AA, and κ scores than all the compared methods on all datasets. This is due to the fact that the proposed MFCT has strong robustness in joint global and local spatial feature extraction and can extract highly discriminative features using larger patches. Combined with the GBiLSTM network that incorporates spectral features, the classification performance can be further improved.

On the IN dataset, for the classes with few samples available, such as classes 1 and 9, our proposed GBiLSTM-MFCT obtains 99.00% and 100% accuracies, which only use three training samples, respectively. The accuracies are significantly higher than other methods. For the UP dataset with sufficient

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TABLE V
CLASSIFICATION ACCURACY COMPARISON FOR THE IN DATASET

Class No.	3DCNN	M3DCNN	SSUN	SSRN	HybridSN	MSOCtCaps	A^2S^2K-ResNet	SpectralFormer		Proposed		
								pixel-wise	patch-wise	GBiLSTM	MFCT	GBiLSTM-MFCT
1	52.00	37.50	86.00	73.13	94.15	95.39	91.67	46.00	43.00	40.00	**99.50**	99.00
2	68.79	60.01	85.48	89.55	90.25	89.99	90.47	51.95	63.46	68.60	93.26	**94.52**
3	32.00	44.85	77.56	86.43	92.90	91.88	93.26	53.18	58.72	46.41	90.77	**94.10**
4	36.92	25.16	58.37	95.55	79.46	88.12	93.48	47.60	34.84	36.92	97.65	**99.82**
5	50.60	41.37	77.97	96.09	87.46	89.57	**98.28**	52.76	56.95	66.23	89.71	93.51
6	89.80	85.13	93.85	**98.14**	96.44	97.68	97.55	87.03	81.34	95.71	96.24	98.05
7	90.91	71.82	90.00	77.97	94.17	**100.00**	77.89	74.55	49.09	70.00	98.18	97.27
8	92.19	97.50	98.84	98.66	99.60	99.27	99.73	90.63	94.64	94.69	**100.00**	99.96
9	87.14	90.00	90.00	97.65	90.00	88.24	61.80	44.29	61.43	68.57	**100.00**	**100.00**
10	64.32	61.58	83.53	91.98	91.12	89.78	93.06	58.33	69.78	55.81	91.60	**94.61**
11	79.64	80.15	92.92	89.68	94.48	86.74	94.03	79.35	75.66	74.04	95.26	**97.04**
12	35.62	39.50	77.74	93.14	79.86	80.87	92.37	29.69	39.25	44.85	89.84	**93.82**
13	95.92	91.10	94.76	99.08	94.35	**100.00**	97.85	86.39	76.96	98.01	96.75	99.79
14	94.03	93.96	95.11	95.63	97.29	95.22	95.67	94.58	85.58	95.05	97.39	**99.28**
15	53.15	44.14	88.84	93.49	88.56	96.36	90.93	18.18	40.77	36.46	95.08	**98.18**
16	86.67	83.91	57.01	**95.76**	77.70	93.78	93.46	80.00	67.82	77.01	86.21	90.11
OA(%)	70.30±1.96	68.85±2.85	87.50±1.04	92.15±1.29	92.26±0.95	90.94±1.51	93.84±0.93	67.14±1.56	69.03±1.13	70.43±0.81	94.28±0.60	**96.46±0.52**
AA(%)	69.36±1.99	65.48±3.44	84.25±0.83	92.00±4.11	90.49±2.33	92.68±1.24	91.34±0.92	62.16±3.78	62.45±2.61	66.77±0.44	94.84±0.55	**96.82±0.77**
$\kappa\times100$	65.65±2.39	64.07±3.32	85.67±1.21	91.03±1.47	91.17±1.09	89.75±1.69	92.97±1.05	62.18±1.76	64.65±1.31	65.96±0.98	93.48±0.69	**95.97±0.59**

TABLE VI
CLASSIFICATION ACCURACY COMPARISON FOR THE UP DATASET

Class No.	3DCNN	M3DCNN	SSUN	SSRN	HybridSN	MSOCtCaps	A^2S^2K-ResNet	SpectralFormer		Proposed		
								pixel-wise	patch-wise	GBiLSTM	MFCT	GBiLSTM-MFCT
1	92.94	95.59	95.02	91.00	96.80	97.49	94.54	83.98	83.84	87.65	98.99	**99.14**
2	95.89	95.15	99.65	97.60	**99.95**	99.71	98.44	95.33	93.76	96.04	99.61	99.41
3	82.07	78.42	63.99	77.73	89.65	94.39	86.44	22.61	41.01	59.58	**97.95**	97.85
4	87.97	94.77	96.32	99.62	95.18	97.14	**99.67**	85.05	91.21	86.46	92.52	98.34
5	96.93	99.48	99.23	99.94	99.48	100.00	98.89	98.68	99.24	99.03	99.85	**100.00**
6	69.11	75.58	96.94	96.27	99.22	99.63	97.91	38.11	54.46	76.55	99.49	**99.66**
7	77.25	77.76	79.29	92.47	92.58	96.44	95.89	70.58	65.98	61.47	96.16	**98.08**
8	90.35	87.18	92.37	84.93	95.02	98.88	81.62	82.42	77.93	89.71	98.41	**99.47**
9	94.54	99.65	88.00	96.30	84.62	96.20	99.04	99.83	97.17	99.18	90.72	**99.65**
OA(%)	89.99±0.80	91.08±1.51	95.09±0.28	93.96±2.69	97.52±0.30	98.66±0.20	95.64±0.63	80.86±1.04	82.86±1.54	88.52±0.43	98.51±0.35	**99.24±0.18**
AA(%)	87.45±1.64	89.29±1.52	90.09±0.97	92.87±2.76	94.72±1.56	97.76±0.55	94.72±1.20	75.18±1.42	78.31±1.67	83.96±1.45	97.08±1.00	**99.07±0.46**
$\kappa\times100$	86.57±1.14	88.12±2.03	93.48±0.39	91.96±3.60	96.71±0.40	98.23±0.26	94.21±0.84	73.95±1.46	76.95±2.24	84.65±0.58	98.03±0.46	**98.99±0.24**

TABLE VII
CLASSIFICATION ACCURACY COMPARISON FOR THE KSC DATASET

CLass No.	3DCNN	M3DCNN	SSUN	SSRN	HybridSN	MSOCtCaps	A^2S^2K-ResNet	SpectralFormer		Proposed		
								pixel-wise	patch-wise	GBiLSTM	MFCT	GBiLSTM-MFCT
1	96.08	96.17	98.57	**100.00**	98.63	97.73	99.72	92.59	98.34	93.85	**100.00**	**100.00**
2	88.61	85.47	88.97	**99.62**	80.99	76.05	98.53	85.47	77.94	89.87	96.95	97.31
3	91.62	91.03	97.78	93.10	93.25	97.71	99.25	75.73	67.26	85.47	**100.00**	**100.00**
4	56.26	49.65	88.70	92.06	79.22	72.86	**99.29**	32.17	39.65	60.26	95.13	96.61
5	45.99	37.96	85.71	91.68	84.76	**94.80**	87.83	22.86	37.01	51.02	93.06	93.20
6	69.09	73.88	86.22	95.57	92.73	89.77	99.72	41.24	70.24	57.51	**100.00**	**100.00**
7	96.42	90.32	99.58	84.14	99.16	**100.00**	93.08	78.68	79.58	71.37	**100.00**	**100.00**
8	89.97	77.16	90.94	98.77	98.18	99.66	99.10	68.15	69.77	88.66	99.14	**100.00**
9	96.36	94.85	98.33	99.66	**100.00**	96.11	**100.00**	77.95	86.78	94.98	**100.00**	**100.00**
10	92.59	95.73	92.81	**100.00**	98.43	95.12	99.95	73.89	88.81	93.51	**100.00**	**100.00**
11	97.25	98.86	99.69	99.64	**100.00**	99.20	99.34	92.88	99.03	96.83	**100.00**	99.90
12	89.76	93.32	98.57	98.68	99.74	97.89	99.23	81.74	91.15	90.63	**100.00**	**100.00**
13	99.91	**100.00**	**100.00**	**100.00**	**100.00**	99.37	**100.00**	99.93	**100.00**	99.91	**100.00**	**100.00**
OA(%)	90.30±1.20	89.18±2.33	95.93±0.79	97.80±0.82	96.48±1.46	95.35±1.16	98.95±0.35	78.89±1.51	84.97±1.73	88.95±0.66	99.34±0.60	**99.49±0.25**
AA(%)	85.38±2.34	83.41±3.65	94.30±0.92	96.38±1.08	94.24±1.91	93.56±1.55	98.08±0.77	69.81±3.20	77.37±2.50	82.60±0.76	98.79±1.19	**99.00±0.56**
$\kappa\times100$	89.19±1.34	87.95±2.60	95.47±0.88	97.55±0.91	96.08±1.62	94.82±1.30	98.83±0.39	76.45±1.71	83.25±1.9	87.68±0.73	99.26±0.67	**99.44±0.28**

samples available, the proposed method obtains an accuracy of higher than 97% for each class. Compared with HybridSN and MSOctCaps, SSRN and A^2S^2K-ResNet have lower classification accuracy on the UP dataset. This might be because smaller input patches lead to poor performance on the UP dataset. On the KSC dataset, the proposed GBiLSTM-MFCT obtains 100% classification accuracy on most of the classes (classes 1, 3, 6–10, 12, and 13) using 4% of the training samples and achieves no less than 93% on the remaining classes.

The second set of experiments is to evaluate the classification performance of our proposed GBiLSTM-MFCT using different training proportion samples. For the IN and KSC

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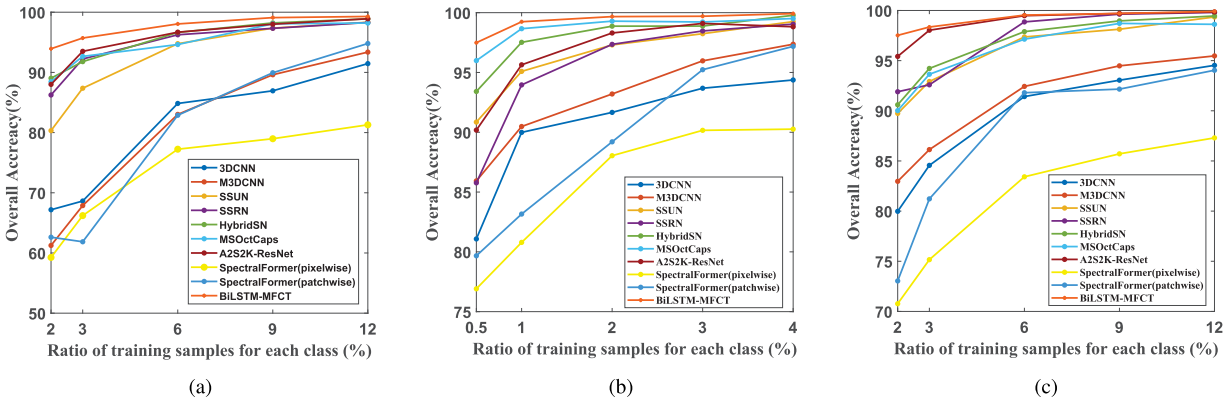


Fig. 5. OA (%) of different methods with different training sample ratios. (a) IN. (b) UP. (c) KSC.

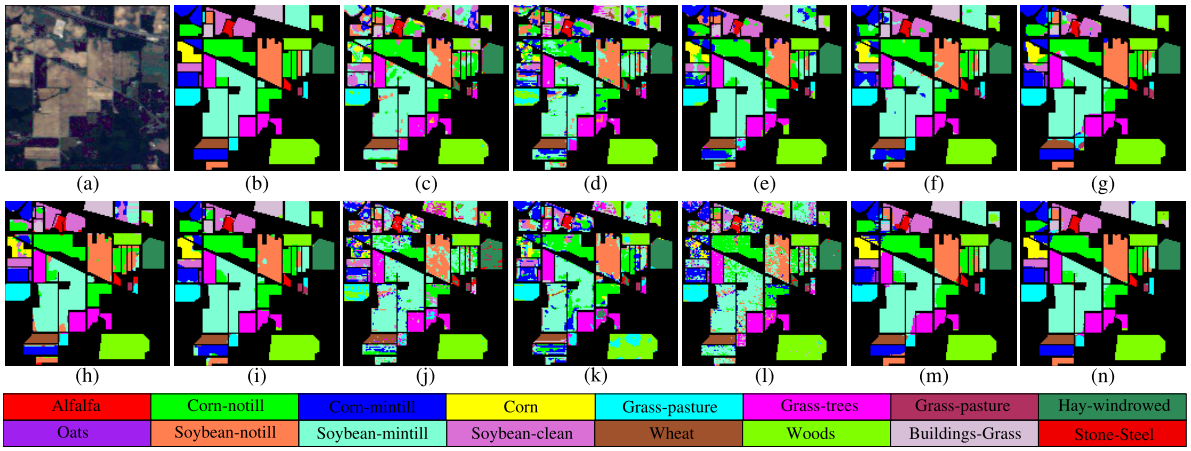


Fig. 6. HSI classification maps of the IN dataset. (a) False-color image. (b) Ground-truth map. (c) 3DCNN. (d) M3DCNN. (e) SSUN. (f) SSRN. (g) HybridSN. (h) MSOctCaps. (i) A^2S^2K-ResNet. (j) SpectralFormer (pixelwise). (k) SpectralFormer (patchwise). (l) GBiLSTM. (m) MFCT. (n) GBiLSTM-MFCT.

datasets, we randomly selected 2%, 3%, 6%, 9%, and 12% samples in each class as training samples. For the UP dataset, we randomly selected 0.5%, 1%, 2%, 3%, and 4% samples in each class as training samples. In these experiments, we randomly selected the same number of samples for each class as the validation set. Each experiment is repeated five times, and Fig. 5 shows the average OA values for different methods with different training percentage samples. It can be found that our proposed GBiLSTM-MFCT outperforms other methods on all datasets, especially for the classes with training samples less than 5% (IN and KSC) and training samples less than 1%(UP).

2) Qualitative Comparisons: In addition to quantitative comparisons, qualitative comparisons for all methods are also provided. Figs. 6–8 show the visualized classification maps of the three datasets, respectively, where (a) is the false color image, (b) is the ground-truth image, (c)–(k) correspond to the classification maps of the eight compared methods, and (l)–(n) correspond to the classification maps of our proposed methods. Different colors in these maps correspond to different categories. From the classification maps, we can see that, compared to our proposed GBiLSTM-MFCT, other methods have more misclassified pixels in the boundary regions and classes that have noise points. This is mainly due to the promising ability of the proposed MFCT for extracting the global–local spatial features that enable the boundary pixels to learn the features of similar pixels close to them.

D. Ablation Study

To further validate the effectiveness of each component proposed in the network, we conduct ablation experiments using the optimal experimental settings (as described in Section IV-B). The baseline methods are BiLSTM and CVT, and then, the proposed components are gradually added to the baseline methods. The compared methods are given as follows.

1) *BiLSTM:* The baseline method for the spectral feature extraction.
2) *GBiLSTM:* The proposed GBiLSTM that is BiLSTM with the grouping strategy.
3) *CVT:* The baseline method for the spatial feature extraction.
4) *CVT + DCE:* CVT using DCE as the convolutional projection embedding layer.
5) *CVT + DCE + BSCA:* CVT with DCE and BSCA.
6) *MFCT:* The proposed multistage fused convolutional transformer for spatial feature extraction.
7) *GBiLSTM-MFCT:* The proposed GBiLSTM-MFCT.

Table VIII shows the ablation analysis of different combinations of modules of the proposed network on the three datasets.

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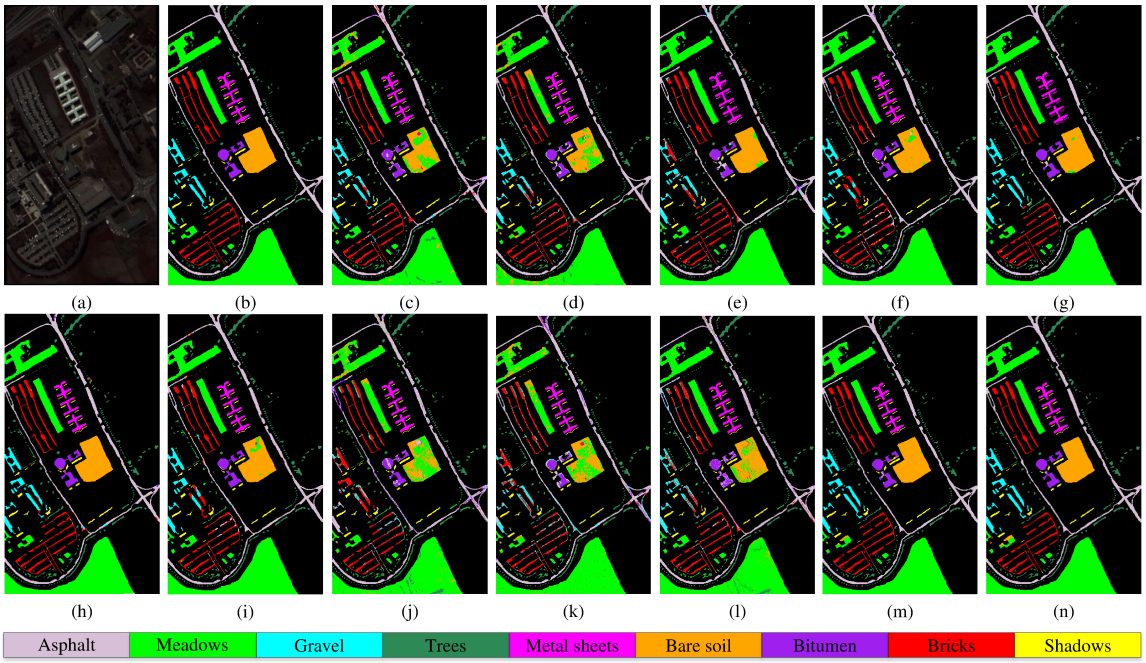


Fig. 7. HSI classification maps of the UP dataset. (a) False-color image. (b) Ground-truth map. (c) 3DCNN. (d) M3DCNN. (e) SSUN. (f) SSRN. (g) HybridSN. (h) MSOctCaps. (i) A^2S^2K-ResNet. (j) SpectralFormer (pixelwise). (k) SpectralFormer (patchwise). (l) GBiLSTM. (m) MFCT. (n) GBiLSTM-MFCT.

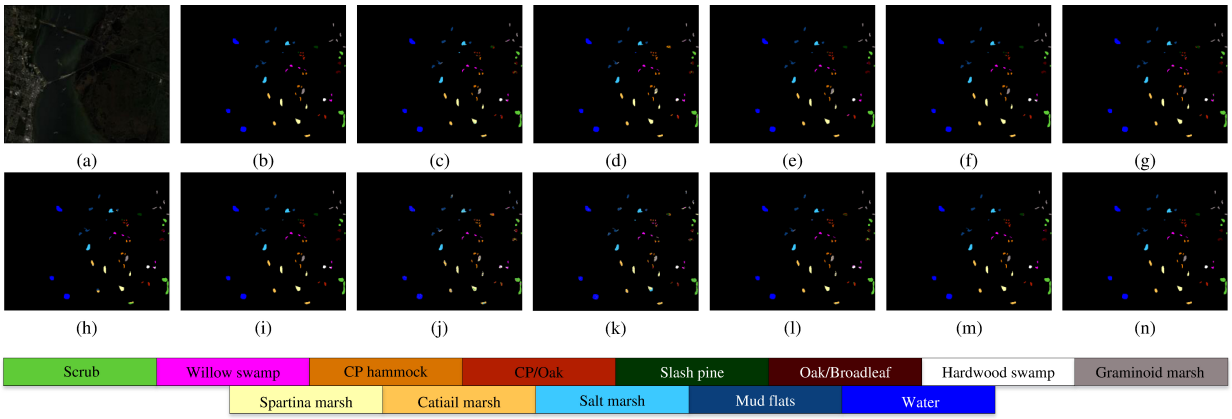


Fig. 8. HSI classification maps of the KSC dataset. (a) False-color image. (b) Ground-truth map. (c) 3DCNN. (d) M3DCNN. (e) SSUN. (f) SSRN. (g) HybridSN. (h) MSOctCaps. (i) A^2S^2K-ResNet. (j) SpectralFormer (pixelwise). (k) SpectralFormer (patchwise). (l) GBiLSTM. (m) MFCT. (n) GBiLSTM-MFCT.

TABLE VIII

ABLATION ANALYSIS OF DIFFERENT COMBINATIONS OF MODULES OF THE PROPOSED FRAMEWORK

Datasets	Matrices	BiLSTM	GBiLSTM	CVT	CVT+DCE	CVT+DCE+BSCA	MFCT	GBiLSTM-MFCT
IN	OA	69.85±0.58	70.43±0.81	90.98±0.85	91.93±0.88	94.07±1.11	94.28±0.60	96.46±0.52
	AA	67.13±0.56	66.77±0.44	90.86±1.02	91.80±1.16	92.79±1.05	94.84±0.55	96.82±0.77
	$\kappa\times100$	65.28±0.77	65.96±0.98	89.73±0.97	90.81±1.00	93.24±1.27	93.48±0.69	95.97±0.59
UP	OA	82.60±0.98	88.52±0.43	96.79±0.25	97.78±0.51	98.34±0.30	98.51±0.35	99.24±0.18
	AA	77.33±1.76	83.96±1.45	93.31±1.09	96.05±1.04	96.90±1.16	97.08±1.00	99.07±0.46
	$\kappa\times100$	76.38±1.48	84.65±0.58	95.74±0.33	97.06±0.69	97.79±0.40	98.03±0.46	98.99±0.24
KSC	OA	79.78±1.27	88.95±0.66	99.05±0.49	99.24±0.54	99.33±0.58	99.34±0.60	99.49±0.25
	AA	72.41±1.72	82.60±0.76	98.33±0.87	98.53±0.90	98.74±1.11	98.79±1.19	99.00±0.56
	$\kappa\times100$	77.45±1.40	87.68±0.73	98.94±0.54	99.15±0.60	99.25±0.64	99.26±0.67	99.44±0.28

From the table, our proposed GBiLSTM improves the OA, AA, and κ greatly on the UP and KSC datasets, and provides a competitive accuracy on the IN dataset in comparison with the BiLSTM. For the extraction of spatial features, we can observe that each proposed component contributes to the improvement of classification performance. The addition of

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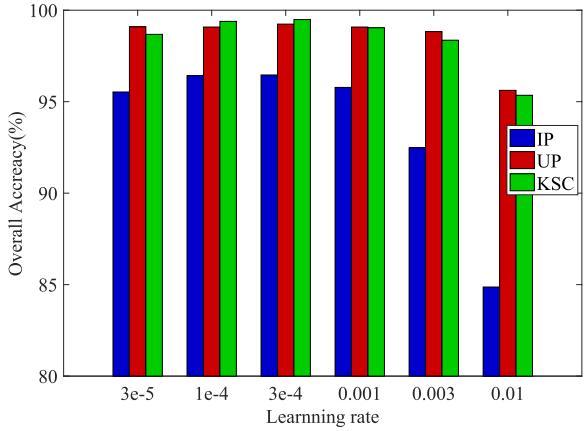


Fig. 9. OA (%) of GBiLSTM-MFCT with different learning rates in the IN, UP, and KSC datasets.

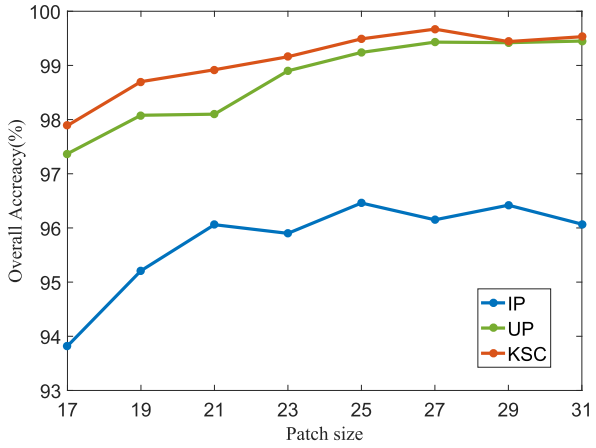


Fig. 10. OA (%) of GBiLSTM-MFCT for different patch sizes in the IN, UP, and KSC datasets.

the DCE and BSCA components both boosts the performance of the IN, UP, and KSC datasets. This can be explained by the fact that the enhancement of the perceptual field brought by DCE contributes to the learning of spatial information. The adopted BSConv that reduces the kernel redundancy enables the BSCA to learn the global and local spatial information more efficiently and, thus, focus on more useful HSI regions. Moreover, we also observe that the multistage fusion strategy leads to an improvement in the classification accuracy from the results of MFCT. This illustrates that the complementary characteristics of the spatial information obtained from the different stages of the MFCT have been fully utilized. Overall, each contributed component of the proposed method is effective. The GBiLSTM-MFCT has the most advanced classification performance compared with the GBiLSTM and MFCT, which verifies that combining spectral and spatial information is more effective than only using spectral or spatial information for HSI classification.

E. Effects of Model Parameters

In the section, the effects of the important parameters of the proposed model GBiLSTM-MFCT, including the learning rate, the image input size of MFCT, and the number of principal components in the preprocessing of PCA, are analyzed.

1) Learning Rate: The learning rate determines the convergence speed and convergence of the model during the training process. Fig. 9 shows a qualitative comparison of the classification performance of different learning rates on the three datasets; we can observe that there is a significant decrease in classification performance when the learning rates are too large. This is because the model would crossover the global optimum during the gradient descent process during training, while the learning rates that are too small would slow down the convergence speed and cause the model to fall into local optimization easily. The results show that the best classification accuracies of IN (96.46%), UP (99.24%), and KSC (99.49%) are obtained at a learning rate of $3e-4$. Considering the aforementioned facts, our learning rate in all the experiments is set to $3e-4$.

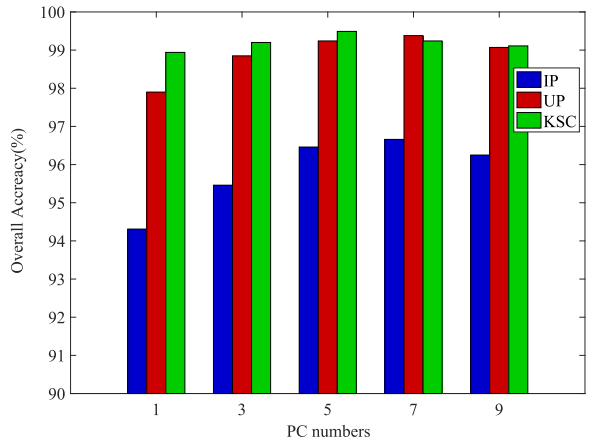


Fig. 11. OA (%) of GBiLSTM-MFCT with different numbers of principal components in the IN, UP, and KSC datasets.

2) Patch Size: The input size of the image determines how much spatial information is contained. Generally, a larger spatial size offers a better classification performance. Fig. 10 shows the OA of different patch sizes on the IN, UP, and KSC datasets. We can see that smaller patches yield low accuracy because smaller patches contain less information, and too large patches do not improve the accuracy greatly due to the excessive smoothing and useless information involved. Therefore, taking the need for sufficient spatial information into account, and in order to avoid the negative effects of too large patches, our model uses a 25×25 patch.

3) Number of Principal Components: Fig. 11 evaluates the impact of the number of principal components on the classification performance of the model. In our experiments, we can see that the classification accuracy of the three datasets tends to level off as the number of principal components increases. A too small number of principal components inevitably bring information loss and, thus, reduce the classification accuracy, and a large amount of useful information is mainly concentrated in the first few principal components. The experimental analysis shows that five spectral dimensions enable our model to achieve good performance in terms of accuracy and computation.

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V. Conclusion

In this article, a novel two-branch network for HSI classification has been introduced. In this architecture, we develop a GBiLSTM network that has significantly reduced the number of parameters required to extract the spectral feature efficiently. The proposed GBiLSTM uses the bidirectional LSTM network and groups the hidden units and the input sequence features, which not only effectively learns the forward and backward spectral information but also reduces the number of parameters that need to be trained for the network. The proposed MFCT, which combines the convolution and transformer, extracts robust global and local spatial features of HSI. The shallow features and deep features obtained from multistage are fused to generate complementary features and further improve the classification performance. In addition, using BSConv as the projection layer of the self-attention module, which focuses on kernel relevance, effectively improves the BSCA module to extract global and local spatial features. The spectral and spatial features extracted by the GBiLSTM network and MFCT are finally fused for classification. The comparison experiments on three public datasets show that our proposed method outperforms all other methods, which verifies that the proposed method can extract more discriminative features than the state-of-the-art methods. In the future, we attempt to construct a new transformer architecture and introduce new attention mechanisms for HSI classification, which can extract the spectral and spatial features of HSI simultaneously.

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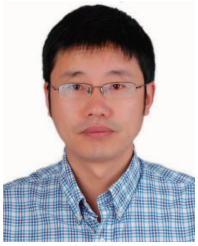
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